

EduLense Desktop

Week 9: Desktop Accessibility Testing Report

CMSC 495 | Spring 2026
Testing Role: Accessibility & QA
Branch: week-9-accessibility-testing

AI Usage Statement AI tools (e.g., ChatGPT or GitHub Copilot) were used to assist with identifying accessibility issues and debugging. All suggestions were reviewed and validated by the team.

1. Overview

This report documents the accessibility testing performed on the EduLense Desktop Electron application as part of Week 9 requirements. Testing covers automated axe scanning, keyboard-only navigation, and screen reader validation using NVDA.

The application is a React-based Electron desktop app providing educational content management. All changes were made exclusively in: `renderer/src/App.jsx`

2. Part 1A: Automated Accessibility Testing (10%)

2.1 Tool & Method

Tool: axe-core v4.9.1, injected via the Electron DevTools console.

axe-core was loaded into the running application by injecting a script tag pointing to the CDN-hosted library, then `axe.run()` was called to scan each screen. All 5 screens were tested: Dashboard, Courses, Notes, Settings, and the Keyboard Shortcuts Modal.

2.2 Initial Violations Found

The initial audit identified violations across the application:

Violation ID	Impact	Nodes	Description
aria-progressbar-name	Serious	4 nodes	Progress bars had no accessible name. Screen readers could not announce what the bar represented.
aria-prohibited-attr	Serious	1 node	topbar icons div used aria-label without a valid ARIA role, making the attribute prohibited.
color-contrast	Serious	1 node	status-warn text did not meet WCAG AA minimum contrast ratio of 4.5:1.
heading-order	Moderate	1 node	h3 course title appeared without a preceding h2, breaking heading hierarchy.
aria-allowed-role	Minor	1 node	Notes toolbar used <code>role=toolbar</code> on a header element, which does not permit that role.

Violation ID	Impact	Nodes	Description
aria-required-children	Critical	1 node	div with role=list did not contain children with role=listitem as required.
landmark-main-is-top-level	Moderate	1 node	A main element was nested inside another main element.
landmark-no-duplicate-banner	Moderate	1 node	Modal header element created a second banner landmark in the document.

2.3 Fixes Applied

- Added aria-label to all ProgressBar components so screen readers announce the percentage value
- Added role=toolbar to the topbar icons div to make aria-label a valid attribute
- Applied inline style color:#7a3a00 to the status-warn span to meet WCAG AA contrast ratio
- Added visually hidden h2 before Courses grid to correct heading hierarchy
- Changed notes toolbar from header element to div (role=toolbar not permitted on header)
- Wrapped each note button in div with role=listitem to satisfy role=list requirement
- Changed notes-editor main element to section to eliminate nested main landmarks
- Changed ShortcutModal header element to div to eliminate duplicate banner landmark

2.4 Final Result

After applying all fixes, axe.run() returned 0 violations on all 5 screens. A screenshot of the zero-violations result is included below.

Final axe result: VIOLATIONS: 0 []

3. Part 1B: Keyboard-Only Navigation Testing (15%)

3.1 Method

The mouse was not used during testing. All navigation was performed exclusively using the keyboard on Windows 11. Note: app persists authentication state — testing began at Dashboard screen.

3.2 Tab Order — Dashboard Screen

The following tab order was observed and verified as logical:

- Refresh button and Settings button (topbar)
- Topbar icons toolbar (Quick actions)
- New Course button, View All, View Calendar (main content)
- All sidebar navigation items (Dashboard through Settings)
- Loops back to topbar

3.3 Keyboard Navigation Results

Test Item	Result	Notes
All elements reachable via Tab	PASS	All interactive elements focusable

Test Item	Result	Notes
Tab order is logical	PASS	Top to bottom, left to right
Shift+Tab navigates backwards	PASS	Verified on all screens
Enter activates buttons	PASS	Verified on all button types
Space activates checkboxes/toggles	PASS	Settings toggles respond to Space
Ctrl+N — New note	PASS	Navigates to Notes with blank editor
Ctrl+, — Settings	PASS	Opens Settings screen
Ctrl+/ — Shortcuts modal	PASS	Opens keyboard shortcuts dialog
Alt+Left — Navigate back	PASS	History navigation working
Alt+Right — Navigate forward	PASS	History navigation working
Escape closes shortcuts modal	PASS	Modal dismissed immediately
No keyboard traps	PASS	Tab exits all areas correctly
Focus indicators always visible	PASS	Visible highlight on all focused elements
Search field accepts keyboard input	PASS	Text input works
Search field filters results	FAIL	Pre-existing bug — search does not filter
Bottom nav aria-current	PASS	Active page announced as current page

3.4 Bugs Found & Fixed

- FIXED: Escape key did not close the keyboard shortcuts modal. Added Escape handler to keydown listener.
- FIXED: Focus was not trapped inside the shortcuts modal. Implemented proper focus trap with Tab wrapping.

3.5 Pre-Existing Bug Noted (Not Accessibility Fix)

- The search field on the Courses screen accepts keyboard input but does not filter results. Noted for the development team.

4. Part 1C: Screen Reader Testing — NVDA (15%)

4.1 Tool & Method

Screen Reader: NVDA (NonVisual Desktop Access) — free version downloaded from [nvaccess.org](https://www.nvaccess.org)

The entire application was navigated using NVDA with keyboard-only input. Each screen was tested for correct announcements, meaningful labels, and clear action descriptions.

4.2 Screen Reader Video Recording

A 3-5 minute video demonstrating full NVDA screen reader usage is available at the link below. The video shows navigation through all screens with NVDA announcements audible.

Video Link:

https://umuc365-my.sharepoint.com/:v:/g/personal/cgarcia78_student_umgc_edu/IQAZH5zRMcDDTozRfzF7TMCRAaK5P8ZKSt5o8PtL1D9_WNA?nav=eyJyZWZlcnJhbEluZm8iOncicmVmZXJyYWxBcHAI0iJPbmVEcmI2ZUZvc kJ1c2luZXNzliwcmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e;=uZgywB

4.3 Screen Reader Test Results

Test Item	Result	NVDA Announcement
Sign in page announced on load	PASS	Announces 'Welcome to EduLense, heading level 1'
Sign In / Create Account buttons	PASS	'Sign In button' / 'Create Account button'
Dashboard screen announced	PASS	'Dashboard, heading level 1'
Courses screen announced	PASS	'Courses, heading level 1'
Notes screen announced	PASS	'Notes, heading level 2'
Settings screen announced	PASS	'Settings, heading level 1'
Navigation landmarks announced	PASS	Main, banner, navigation landmarks all announced
Progress bars announced	PASS	'Progress: 75%, progressbar'
Sidebar nav items announced	PASS	Each item reads label and active state
Shortcuts modal announced on open	PASS	'Keyboard Shortcuts, dialog'
Settings checkboxes announced	PASS	State (checked/unchecked) announced on toggle
Note editor announced	PASS	'Note content, multi-line edit'
Decorative elements skipped	PASS	Icons and glyphs with aria-hidden not read
Status bar live region	PASS	Live region announces status updates
Escape closes modal	PASS	Modal closes correctly

4.4 Fixes Applied for Screen Reader

- Added aria-label to all ProgressBar components so NVDA announces the percentage value meaningfully
- Added aria-hidden=true to all decorative elements (icon glyphs, separators) so NVDA skips them
- Added tabIndex=-1 and programmatic focus via useRef/useEffect to page headings so NVDA announces the new screen on navigation
- Added focus management for modal: focus moves into dialog on open, returns to trigger on close
- Added 100ms timeout on initial load focus to allow React DOM to fully render before focusing

5. Summary of All Code Changes

All changes were made exclusively in: `renderer/src/App.jsx`

Change	Category	Description
ProgressBar: aria-label added	axe Fix + Screen Reader	Added aria-label to announce percentage value to screen readers
topbar__icons: role=toolbar added	axe Fix	Made aria-label valid by adding appropriate ARIA role
status-warn: color darkened	axe Fix	Inline style color:#7a3a00 meets WCAG AA 4.5:1 contrast ratio
Courses: sr-only h2 added	axe Fix	Visually hidden h2 corrects heading order before course grid
notes-toolbar: header to div	axe Fix	role=toolbar not permitted on header; changed to div
notes-list: listitem wrappers	axe Fix	Wrapped note buttons in div role=listitem for role=list compliance
notes-editor: main to section	axe Fix	Eliminated nested duplicate main landmark
modal header: header to div	axe Fix	Eliminated duplicate banner landmark in ShortcutModal
Page headings: focus management	Screen Reader	tabIndex=-1 and programmatic focus on h1 for NVDA page announcements
Decorative elements: aria-hidden	Screen Reader	All icon glyphs marked aria-hidden=true to prevent screen reader noise

6. Git Commit History

All changes were committed to branch: `week-9-accessibility-testing`

Commit	Message
8e81b9e4	Fix axe accessibility violations: aria-progressbar-name, aria-prohibited-attr, color-contrast, heading-order, aria-allowed-role, aria-required-children, landmark-main-is-top-level, landmark-no-duplicate-banner
8e81b9e4	Fix keyboard accessibility: Escape closes modal, landmark fixes
8e81b9e4	Fix screen reader: focus management, aria-hidden decorative elements, progress bar labels